# Omar Khan

Ph.D. Candidate in Human-AI Interaction

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## Summary

Mixed-methods HCI researcher working across human-centered AI, accessibility, and data visualization. I build systems, most recently QUARTZ, for accessible qualitative analysis that let blind and low-vision researchers analyze their own data and author their own visualizations independently. Designing with BLV scientists surfaces methods that transfer beyond disability contexts to STEM education and collaborative analysis more broadly.

## Education

### University of Illinois Urbana-Champaign

Ph.D., Computer Science 08/2022 – 05/2028

***Thesis:*** *Toward Equitable Research: Co-Designing Accessible and Intelligent Research Workflows with Blind and Low-Vision Researchers*
***Advisor:*** *Dr. JooYoung Seo*

*Certificate in Information Accessibility Design and Policy*

### University of Illinois Urbana-Champaign

B.S., Computer Science + Crop Sciences 2017 – 2021

## Research Experience

### (x)Ability Design Lab

*Champaign, IL*

Graduate Research Assistant | Advisor: Dr. JooYoung Seo 05/2024 – Present

- **QUARTZ (Qualitative Understanding via Accessible Representation and Visualization)**
  - Led user research to understand barriers in qualitative data visualization, conducting **8** co-design sessions via the **RITE method**
  - Designed user study protocols, including survey instruments, semi-structured interview protocols, and task-based usability testing
  - Managed global participant recruitment pipeline via BLV advocacy networks (e.g., World Blind Union)
- **Re-Imagining Knowledge Production Workflows by, with, and for BLV Researchers**
  - Led a mixed-methods global study with BLV researchers (survey = 57; interviews = 15) using activity theory to examine research tool barriers, revealing **17% task delegation rate** and critical gaps in data visualization tools **(2.58/5 difficulty)**.
  - Synthesized findings through reflexive thematic analysis to develop systemic design recommendations addressing workflow contradictions across literature review, analysis, and dissemination stages.
- **Parameter-Efficient Fine-Tuning of LLMs for Accessibility**
  - Fine-tuned Gemma 3 1B-IT with LoRA/PEFT on the NCSA Delta cluster for web accessibility and WCAG question answering, on a purpose-built instruction dataset authored for the task
  - Trained a Flan-T5-Large LoRA adapter for disability question answering and rewriting of ableist language, targeting bias mitigation in generated text
- **Accessible Extended Reality (XR) Co-Design**
  - Co-designed *PunchPulse*, an accessible VR boxing game with a **5-person mixed-abilities team**, leveraging the Rapid Iterative Testing and Evaluation (RITE) framework over 16 weeks
- **Multimodal LLM-Based Data Visualization**
  - Contributed to MAIDR, a multimodal AI-driven system for accessible data visualization, as part of a team of **8 developers** and **3 researchers**.

### CROPS Lab

*Urbana, IL*

Graduate Research Assistant | Advisor: Dr. Camille Cobb 08/2022 – 05/2024

- **YouTube Influencer Ads and Privacy Promises**
  - Led large-scale quantitative analysis of **300M YouTube influencer privacy technology advertisements**, identifying top-advertised products and consumer misinformation risks.
- **Smart Home Data Collection Consent Dynamics**
  - Co-led mixed-methods analysis of **360-participant survey** examining consent dynamics in smart home environments, revealing **84% of users experienced consent violations**.
  - Developed stakeholder-specific recommendations for consentful smart home ecosystems.

### National Center for Supercomputing Applications
*Urbana, IL*

Research Assistant | Advisors: Dr. Megan Dailey & Dr. Matthew Hudson — 08/2021 – 12/2021

- **Food Bank Marketplace Mobile App**
  - Developed the initial software design specification for a mobile app aimed to create a marketplace to provide food banks with the means to procure fresh produce from farmers at a reduced cost, aiming to reduce food waste and insecurity.

### ORCHID Lab
*Urbana, IL*

Undergraduate Research Assistant | Advisor: Dr. Brian P. Bailey — 01/2021 – 05/2021

- **Proximal Process Goal Accomplishment**
  - Researched methodologies and visualizations to help individuals accomplish proximal process goals (goals that are achievable in a short amount of time and involve specific processes).

### Siebel School of Computing and Data Science
*Urbana, IL*

Undergraduate Research Assistant | Advisor: Dr. Lawrence C. Angrave — 01/2021 – 05/2021

- **Webcam-Based Eye Tracking in ClassTranscribe**
  - Investigated the challenges of webcam-based eye tracking and an implementation of it in ClassTranscribe, an online lecture streaming platform at the University of Illinois, using WebGazer, a React.js-based framework developed by Brown University.

### Siebel School of Computing and Data Science
*Urbana, IL*

Undergraduate Research Assistant | Advisor: Dr. Geoffrey L. Herman — 05/2019 – 08/2019

- **Computerized Exam for Computer Architecture**
  - Designed, implemented, and deployed a computerized, multi-part exam with another undergraduate student on pipelining and cache analysis for CS 233: Computer Architecture, taken annually by **400+ students**, introducing hardware concepts
  - Co-authored a publication on the implications of our implementation on engineering education beyond computing

## Professional Experience

### American Foundation for the Blind (AFB)
*Remote*

Research Fellow — 11/2026 – Present

- Co-designing a national mixed-methods study on how workplace AI policies affect workers with disabilities

### Infosys
*Remote*

Research Scientist Intern (Center for Autonomous Machines) — 06/2026 – 09/2026

- Led a **convergent mixed-methods study** (four online focus groups, **n=14**, plus follow-up survey) to surface universal design principles for autonomous last-mile delivery robots (LMDRs); thematic analysis of session logs produced **seven design themes** for pedestrian-robot interaction
- Engineered the study stimulus end-to-end: a **Three.js** browser simulation of a pedestrian-robot encounter driven by a finite-state behavior model, with ARIA live regions and reduced-motion support, validated by a headless regression suite
- First author on a technical paper with the Infosys Center for Autonomous Machines, under submission; provides targeted recommendations for inclusive LMDR design

### Fable
*Remote*

Accessibility Consultant (Contract) — 09/2025 – Present

- Conducted **200+** accessibility evaluations, including moderated user sessions, prototype reviews, and assistive technology compatibility testing (screen and OS magnification) -- for enterprise clients across e-commerce, health, and technology
- Auditing against **WCAG 2.2 AA** across U.S., Canadian, and U.K. regulatory contexts (ADA/Section 508, ACA, EAA); delivering severity-ranked remediation guidance to product and engineering teams

### American Foundation for the Blind (AFB)
*Remote*

Researcher (Volunteer) — 09/2025 – 11/2026

- Analyzing a **national survey of 1,735 U.S. adults** (1,070 disabled, 665 non-disabled) to isolate differential experiences with AI systems, a matched-comparison design that separates disability-specific barriers from general usability friction
- Co-author on **three public reports** in AFB's AI series, spanning transportation access, employment, and cross-cutting benefits and risks; findings released through national press coverage and a public policy webinar

### Viasat
*Carlsbad, CA*

Software Engineer Intern — 05/2021 – 08/2021

- Designed and implemented a Slack app with 2 other interns using Python (frontend), SQLAlchemy, and Kubernetes (backend) that allows users to manually archive older Jira projects and automatically archive them by a given date

**L'Oréal U.S.A.** *Remote*

Software Engineer Extern 06/2020 – 08/2020

- Collaborated with a team of 4 externs in a product design competition, implementing a data-driven solution to reduce the amount of plastic used in L'Oréal product packaging

## Publications and Presentations

Conference and Journal Publications

(C1) **Omar Khan** and JooYoung Seo. 2026. QUARTZ: Qualitative Understanding via Accessible Representation and Visualization. To appear in *The 28th International ACM SIGACCESS Conference on Computers and Accessibility (ASSETS '26), October 25–28, 2026, Vila Nova de Gaia, Portugal. ACM.* [Preprint]

(C2) **Omar Khan** and JooYoung Seo. 2026. "I Don't Want My Mental Health App To Give Me Mental Health Barriers": Unpacking The Need For Digital Mental Health Tracking Services With And For The Blind Community. To appear in *The 28th International ACM SIGACCESS Conference on Computers and Accessibility (ASSETS '26), October 25–28, 2026, Vila Nova de Gaia, Portugal. ACM.* [Preprint]

(C3) **Omar Khan** and JooYoung Seo. 2026. "I Don't Trust Any Professional Research Tool": A Re-Imagination of Knowledge Production Workflows by, with, and for Blind and Low-Vision Researchers. *In Proceedings of the 2026 CHI Conference on Human Factors in Computing Systems (CHI '26). April 13–17, 2026, Barcelona, Spain.* ACM, New York, NY, USA, 24 pages. [DOI] [arxiv]

(C4) Sanchita S. Kamath, **Omar Khan**, Anurag Choudhary, Jan Meyerhoff-Liang, Soyoung Choi, JooYoung Seo. 2025. PunchPulse: A Physically Demanding Virtual Reality Boxing Game Designed with, for and by Blind and Low-Vision Players. In *The 27th International ACM SIGACCESS Conference Computers and Accessibility (ASSETS '25)*, October 26–29, 2025, Denver, CO, USA. ACM, New York, NY, USA, 21 pages. [DOI]

(C5) Alex Atcheson, **Omar Khan**, Brian Siemann, Anika Jain, and Karrie Karahalios. 2025. "I'd Never Actually Realized How Big An Impact It Had Until Now": Perspectives of University Students with Disabilities on Generative Artificial Intelligence. In *CHI Conference on Human Factors in Computing Systems (CHI '25), April 26–May 01, 2025, Yokohama, Japan.* ACM, New York, NY, USA, 22 pages. [DOI]

(C6) Yi-Shyuan Chiang, **Omar Khan**, Adam Bates, and Camille Cobb. (2024). More than just informed: The importance of consent facets in smart homes. In *Proceedings of the CHI Conference on Human Factors in Computing Systems (CHI '24), May 11–16, 2024, Honolulu, HI, USA.* ACM, New York, NY, USA, 21 pages. [DOI]

(C7) Suleman Mahmood, Mingjie Zhao, **Omar Khan**, Geoffrey L. Herman. (2020). Caches as an Example of Machine-gradable Exam Questions for Complex Engineering Systems. In *Proceedings of the 49th ASEE/IEEE Frontiers in Education Conference, Uppsala, Sweden, Oct. 21–24.* [DOI]

Public Reports & White Papers

(R1) Carmel H. Heydarian, Angie L. Whistler, Arielle M. Silverman, Sarahelizabeth J. Baguhn, William E. Hanuschock III, Mana Hashimoto, **Omar Khan**, Alyssa Shock, Mei-Lian Vader. (2026). Navigating Toward AI: Transportation Access in a Changing Technology Landscape. *American Foundation for the Blind.* [Link]

(R2) Alyssa Shock, Sarahelizabeth J. Baguhn, William E. Hanuschock III, Mana Hashimoto, Carmel Heydarian, **Omar Khan**, Mei-Lian Vader, Angie L. Whistler. (2026). Working with the Machine: AI's Expanding Role in Employment for People With Disabilities. *American Foundation for the Blind.* [Link]

(R3) Arielle M. Silverman, Angie L. Whistler, Alyssa Shock, Carmel H. Heydarian, Sarahelizabeth J. Baguhn, William E. Hanuschock III, Mana Hashimoto, **Omar Khan**, & Mei-Lian Vader. (2026). The AI Quagmire: Benefits, Risks, and User Aspirations Through a Disability Lens. *American Foundation for the Blind.* [Link]

Works in Progress, Posters, and Presentations

(W1) **Omar Khan** and JooYoung Seo. 2025. "Sighted People Have Their Pick Of The Litter": Unpacking The Need For Digital Mental Health (DMH) Tracking Services With And For The Blind Community. In *Extended Abstracts of the CHI Conference on Human Factors in Computing Systems (CHI EA '25), April 26–May 1, 2025, Yokohama, Japan.* ACM, New York, NY, USA, 13 pages. [DOI]

(W2) Sanchita S. Kamath, Aziz Zeidieh, **Omar Khan**, Dhruv Sethi, and JooYoung Seo. 2024. Playing Without Barriers: Crafting Playful and Accessible VR Table-Tennis with and for Blind and Low-Vision Individuals. In *Proceedings of the 26th International ACM SIGACCESS Conference on Computers and Accessibility (ASSETS '24).* Association for Computing Machinery, New York, NY, USA, Article 88, 1–5. [DOI]

(W3) Harsh Deep, Joanna Huang, Rashi Dembi, **Omar Khan**, and Lawrence Angrave. Webcam Eye-Tracking Based Accessibility for ClassTranscribe. *Virtual poster presentation: Illinois Undergraduate Research Symposium 2021.*

(W4) Sanchita S. Kamath, **Omar Khan**, Aziz N. Zeidieh, and JooYoung Seo. 2025. Sensing the Shape of Data: Non-Visual Exploration of Statistical Concepts in Histograms with Blind and Low-Vision Learners. *arXiv preprint arXiv:2509.14452.* [arxiv]

## Service and Leadership

### Research Service

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| 2026 | Technical Paper Reviewer | *DIS 2026* |

### Community Service

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| 2026-Present | Vice President, Board of Directors \| PACE - Center for Independent Living | *Urbana, IL* |
| 2024-2025 | Broadening Participation in Computing (BPC): K-12 Student Outreach Committee | *UIUC* |
| 2023-Present | Director, Board of Directors \| PACE - Center for Independent Living | *Urbana, IL* |
| 2021-2022 | Campus Co-Director, Fundraising Team \| The Percentage Project Inc. | *Remote* |
| 2020-2022 | Facilitator, Curriculum Development Co-Lead \| Girls Who Code | *UIUC* |
| 2021 | Mentoring Co-Chair \| Illinois Women in Computer Science | *UIUC* |
| 2019-2020 | Content Co-Chair \| Reflections \| Projections | *UIUC* |
| 2019-2020 | Co-Director \| Illinois Computer Science Sail | *UIUC* |

## Teaching Experience

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| S24 | Graduate Teaching Assistant: CS 467: Social Visualization | *UIUC* |
| F23, F24 | Graduate Teaching Assistant: CS 210/211: Ethical & Professional Issues/Conduct in CS | *UIUC* |
| F19, S20, F20, S21 | Undergraduate Course Assistant: CS 233: Computer Architecture | *UIUC* |
| F18, S19, F19 | Undergraduate Course Assistant: CS 196-25: Freshman Honors | *UIUC* |
| F18, S19 | Undergraduate Course Assistant: CS 125: Introduction to Computer Science | *UIUC* |

## Mentoring

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| 2025-2026 | Soundarya Kumar Pradhan | *UIUC MCS* |

## Awards and Honors

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| 08/2025 | SIGACCESS Travel Scholarship | *ACM* |
| 12/2024 | CRA-WP Grad Cohort for Women & Grad Cohort for IDEALS (2025 cohort) | *CRA-WP* |
| 04/2021 | Most Long-Term Social Impact: Stanford Open Datathon 2021 | *Stanford* |
| 2020–2021 | The Climate Corporation Scholarship | *UIUC* |

## Skills

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| **Programming Languages** | Python, Java, JavaScript, R, Swift, C++ |
| **Technologies** | HTML, CSS, Git, Figma, Ollama, Qualtrics, Claude Code, Codex, Docker, MAXQDA, SQL, Kubernetes, MongoDB, Neo4j |
| **Frameworks** | React, Next.js, PyTorch, SciPy, TensorFlow, scikit-learn Matplotlib, NumPy, Pandas, NLTK, D3.js, Flask, Hadoop, Spark, SQLAlchemy |
| **Research Methods** | Co-Design, Interview Design, Semi-Structured Interviews, Survey Design, Thematic Analysis Statistical Analysis, A/B Testing, Experimental Design, Prototyping, Usability Testing Affinity Diagrams, Heuristic Evaluation |